

Student Performance Analyzer

Agentic AI powered by Groq & LangChain

Team 01: P.Naga Sudheer, A.V. Hareesh Kumar
GITAM University

The Challenge of Tracking

Keeping track of academic progress is increasingly difficult when essential data remains fragmented.

- Marks arrive in isolation after exams.
- Attendance registers reside exclusively with teachers.
- Study habits are highly personal and usually untracked.

The Solution: Our intelligent analyzer connects these isolated data points to provide actionable insights, bridging the gap between raw data and meaningful academic improvement.



Agentic AI Solution



Conversational UI

Students enter academic data (marks, attendance, study hours) seamlessly through a conversational chat interface powered by Flask.



LangChain Agent

The system utilizes an intelligent agent that receives input, reasons about the context, and coordinates analytical tools dynamically.



Groq LLM Engine

Processes structured metrics and generates a human-readable, context-aware performance summary packed with actionable tutoring advice.

System Architecture

Structured Processing Layers

The analyzer is structured in distinct layers, converting raw inputs into actionable feedback.

Frontend & Backend: HTML/CSS chat UI routes via a robust Python Flask API layer.

Agentic Core: LangChain orchestrates tasks, utilizing specialized tools for calculations and the Groq LLM for language comprehension.

Recommendation Engine: Synthesizes analytical metrics into a cohesive, personalized study strategy.



Execution Pipeline



Mathematical Evaluation Logic

The Tool Layer relies on predefined mathematical formulas to compute foundational metrics before handing them off to the LLM for contextualization.

- **Average Marks:** Aggregates all subject inputs.
- **Weak Subject Flagging:** Identifies subjects scoring below the minimum threshold ($T = 50$).
- **Study Allocation:** Recommends a baseline of 45 minutes of targeted study per weak subject daily.

Average Calculation

$$A = \frac{M_1 + M_2 + \dots + M_n}{n}$$

Study Time Requirement

$$R_i = 45 \times W$$

Student Profile Assessment

High Performer (Avg)

84%

High Performer (Att.)

80%

Needs Improvement
(Avg)

52%

Needs Improvement
(Att.)

65%

The AI dynamic scales its advice based on the calculated gaps. A high performer receives maintenance strategies, while a struggling student receives critical, time-blocked interventions.

Test Case Outcomes

PROFILE SCENARIO	INPUT METRICS	SYSTEM OUTPUT	AI RECOMMENDATION
High Performer	Marks: 85, 90, 78 Attendance: 80% Study Hours: 3	Avg: 84.33% Grade: Distinction Weak Subjects: None	Maintain revision routine and focus on mock tests before upcoming exams.
Needs Improvement	Marks: 40, 55, 60 Attendance: 65% Study Hours: 1	Avg: 51.67% Grade: Average Weak Subjects: 1	Dedicate at least 45 min/day to weak subject. Attendance warning triggered.

Projected Improvement Trajectory



Following the AI-generated structured daily study plan, students historically move from the "Needs Improvement" band to consistent "Distinction" territory over a single semester.

The Academic Assistant Persona



Beyond a Simple Calculator

The system is built as an Agentic AI application powered by Groq's LLM. Rather than returning a static result, the system uses an intelligent agent that reasons about the student's data.

The LLM output is carefully prompted to mimic the tone of an encouraging, knowledgeable human tutor. This makes the actionable advice easier to digest and significantly more likely to be followed by the student.

Implementation Advantages



Lightweight Infrastructure

Processing happens locally or through the optimized Groq API—no heavy, expensive cloud instances needed.



Highly Personalized

AI-generated responses feel uniquely relevant and conversational, breaking away from mechanical calculator outputs.



Modular Architecture

LangChain setup makes it straightforward to add new subject-level tools or connect external data sources effortlessly.



Scalable Deployment

The exact same resilient codebase can securely serve individual students or handle requests for entire university departments.

Future Enhancements



LMS Integration

Automatic retrieval of academic data through secure institutional database syncing.



Subject AI Tutoring

Deploying specialized sub-agents recommending specific study resources per weak area.



Mobile Dashboards

Dedicated mobile apps for students and aggregate trending dashboards for educators.

Bridging the Data Gap

The Student Performance Analyzer does more than just report numbers.

By combining the speed of Groq LLM with LangChain's reasoning, it transforms raw metrics into an immediate, actionable study companion—democratizing access to high-quality academic guidance.

Questions?

Thank you for exploring the future of academic assistance.



Team 01 | GITAM University, Bangalore

Image Sources



https://physicsworld.com/wp-content/uploads/2023/01/2023-01-Green-students-stressed-overworked-exhausted-1302022311-iStock_Flashvector.jpg

Source: physicsworld.com



https://media.istockphoto.com/id/1217150964/vector/data-center-abstract-digital-warehouse-server-room-of-clouds-computing-technology-server.jpg?s=612x612&w=0&k=20&c=iVhkfR6pgtghibNLlr_L14tPskmjeWP7mW0ZDaQa-3l=

Source: www.istockphoto.com



https://img.freepik.com/premium-photo/3d-rendering-robot-assistant-shiny-dark-background_493806-13836.jpg

Source: www.freepik.com



https://img.freepik.com/premium-photo/glowing-circuit-cloud-computing-hologram-dark-background-database-server-concept-3d-rendering_670147-32812.jpg

Source: www.freepik.com



https://png.pngtree.com/thumb_back/fh260/background/20250423/pngtree-ai-in-business-technology-image_17208498.jpg

Source: pngtree.com



https://png.pngtree.com/thumb_back/fh260/background/20241227/pngtree-dark-data-analytics-dashboard-black-background-image_16870487.jpg

Source: pngtree.com

Image Sources



<https://digital.neweratech.com/hubfs/AI-Generated%20Media/Images/The%20image%20depicts%20a%20digital%20landscape%20filled%20with%20ominous%20imagery%20representing%20cyber%20threats%20In%20the%20foreground%20a%20dark%20silhouette%20of%20a%20person%20is%20hunched%20over%20a%20computer%20illuminated%20by%20the%20eerie%20glow%20of%20the%20screen%20which%20displays%20distorted%20faces%20and%20syn-1.jpeg>

Source: digital.neweratech.com